

Proof of conjecture 00000002478: the q-binomial (Gauss) expansion and its finite-field counting content

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2026-10-05

1 The conjecture and its reading

Statement (English, verbatim). Definition: The q-binomial theorem: the Gauss expansion. Conjecture: The q-binomial theorem generalizes to the Gauss expansion, with the counting content given by the lattices of finite fields – the q-binomial coefficient counts subspaces, and the expansion closes under the Gaussian lattice. (q-binomial theorem Gauss expansion finite-field lattice)

Chinese text. It says, in substance, that the q-binomial theorem has a Gauss generalization, and that its combinatorics is given by the lattices of finite fields.

These are classical theorems, not new results. We prove them and check the proofs in Lean 4 with Mathlib. We read the statement as four claims. Each one is a theorem in the Lean file.

(E) *The expansion.* As polynomials in two variables q, t with integer (indeed natural) coefficients,

$$\prod_{i=0}^{n-1} (1 + q^i t) = \sum_{k=0}^n q^{k(k-1)/2} \begin{bmatrix} n \\ k \end{bmatrix}_q t^k.$$

The same identity holds after evaluating q and t in any commutative semiring. Wikipedia [W] states this exact formula and names it the “Cauchy binomial theorem”. The conjecture calls it the Gauss expansion.

(C1) *The coefficient counts subspaces.* Let $\mathbb{F}$ be a finite field with q elements. The number $N_q(n, k)$ of k -dimensional subspaces of $\mathbb{F}^n$ is $\begin{bmatrix} n \\ k \end{bmatrix}_q$ evaluated at q .

(C2) *Counting content of the expansion.* For $q = |\mathbb{F}|$, $\prod_{i < n} (1 + q^i t) = \sum_{k \leq n} q^{k(k-1)/2} N_q(n, k) t^k$ in $\mathbb{N}[t]$.

(P) *The expansion closes.* The coefficients and the counts satisfy the q-Pascal recursion $\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q + q^{k+1} \begin{bmatrix} n \\ k+1 \end{bmatrix}_q$. Multiplying by one more factor turns the order- n expansion into the order- $(n+1)$ expansion exactly by this recursion; see the proof of (E).

Not covered here. We read “closes under the Gaussian lattice” as (P). A different reading is *interval closure*: every interval of the subspace lattice is again counted by Gaussian binomials. That reading belongs to the subspace-count conjecture 00000002490, and it is proved in our separate package for that conjecture. This package does not repeat it.

Conventions. $\mathbb{F}$ is any finite field with $q = |\mathbb{F}| \geq 2$; its prime-power form is not used. A subspace is a `Mathlib Submodule`. Dimension is `finrank`. A count is `Nat.card`. $k(k-1)/2$ is natural-number division, and it equals $\binom{k}{2}$ (`Nat.choose_two_right`). The two-variable polynomial ring is $\mathbb{N}[q][t]$, with q the inner variable and t the outer one.

2 Definitions as formalized

Definition 1 (`gaussBinom`). The Gaussian binomial $\begin{bmatrix} n \\ k \end{bmatrix}_q \in \mathbb{N}[q]$ is defined by the rules $\begin{bmatrix} n \\ 0 \end{bmatrix}_q = 1$, $\begin{bmatrix} 0 \\ k+1 \end{bmatrix}_q = 0$, and $\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q + q^{k+1} \begin{bmatrix} n \\ k+1 \end{bmatrix}_q$.

This is the first “analog of Pascal’s identity” in [W]. Wikipedia instead *defines* the coefficient by the quotient $(1 - q^m) \cdots (1 - q^{m-r+1}) / ((1 - q) \cdots (1 - q^r))$. We prove (`gaussBinom_mul_qDen`) that $\begin{bmatrix} n \\ k \end{bmatrix}_q \prod_{i < k} (1 - q^{i+1}) = \prod_{i < k} (1 - q^{n-i})$ holds in every commutative ring, so the two definitions agree. For $k > n$ both sides are 0, which matches [W] (“If $r > m$, this evaluates to 0”). We also show that $\begin{bmatrix} n \\ k \end{bmatrix}_q = 0$ for $n < k$ (`gaussBinom_eq_zero_of_lt`).

3 Proofs

Theorem 2 ((E), `qBinomial_eval`, `qBinomial_expansion`). *In every commutative semiring, for all x, t , $\prod_{i < n} (1 + x^i t) = \sum_{k \leq n} x^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q (x) t^k$.*

Proof. The proof is by induction on n , for all t at once; $n = 0$ is trivial. Peel off the factor $i = 0$:

$$\prod_{i < n+1} (1 + x^i t) = (1 + t) \prod_{i < n} (1 + x^i (xt)) = (1 + t) \sum_{k \leq n} a_k x^k t^k,$$

where $a_k = x^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q (x)$ and the induction hypothesis is used at xt . Since $\begin{bmatrix} n \\ n+1 \end{bmatrix}_q = 0$, we have $\sum_{k \leq n} a_k x^k t^k = 1 + \sum_{k \leq n} a_{k+1} x^{k+1} t^{k+1}$. So the coefficient of t^{k+1} is $a_{k+1} x^{k+1} + a_k x^k = x^{\binom{k+1}{2}} (\begin{bmatrix} n \\ k \end{bmatrix}_q + x^{k+1} \begin{bmatrix} n \\ k+1 \end{bmatrix}_q) = x^{\binom{k+1}{2}} \begin{bmatrix} n+1 \\ k+1 \end{bmatrix}_q$. Here we used $\binom{k+1}{2} = \binom{k}{2} + k$ and the q-Pascal rule (P). The constant term is 1 on both sides. The polynomial identity follows by evaluating at $x = q$ and t in $\mathbb{N}[q][t]$. $\square$

Theorem 3 ((C1), `card_subspaces_eq_gaussBinom`, `subCount_eq`). *Let V be a finite $\mathbb{F}$ -vector space of dimension n . Then $\#\{W \leq V : \dim W = k\} = \begin{bmatrix} n \\ k \end{bmatrix}_q (q)$.*

Proof. Double count the ordered linearly independent k -tuples. Each one spans a k -dimensional subspace W . The tuples that span a fixed W are exactly the linearly independent k -tuples of W (`fibreEquiv`). `Mathlib`’s `card_linearIndependent` counts the linearly independent k -tuples in an m -dimensional space as $\prod_{i < k} (q^m - q^i)$ when $k \leq m$. This gives $\prod_{i < k} (q^n - q^i) = N \cdot \prod_{i < k} (q^k - q^i)$ for $k \leq n$ (`card_li_eq`). By induction on n via q-Pascal, $\begin{bmatrix} n \\ k \end{bmatrix}_q (x) \prod_{i < k} (x^k - x^i) = \prod_{i < k} (x^n - x^i)$ in every commutative ring (`gaussBinom_mul_liProd`). Since $q \geq 2$, we may cancel $\prod_{i < k} (q^k - q^i) \neq 0$ in $\mathbb{Z}$. For $k > n$, both N and $\begin{bmatrix} n \\ k \end{bmatrix}_q (q)$ are 0. $\square$

(C2) is (E) evaluated at $x = q$, with each $\begin{bmatrix} n \\ k \end{bmatrix}_q (q)$ replaced by $N_q(n, k)$ using (C1) (`qBinomial_expansion_count`). (P) for the counts is the defining recursion, evaluated at q and transported by (C1) (`subCount_pascal`).

4 Lean formalization

The file is `lean/Conjecture2478/Basic.lean`, namespace `C2478`, 304 lines. Its only import is `Mathlib`. The main theorem `conjecture_2478` is a conjunction. It states (E) as an identity in $\mathbb{N}[q][t]$ (Lean `Polynomial (Polynomial Nat)`) for all n . For every finite field $\mathbb{F}$, it also states (C1) for $\text{Fin } n \rightarrow \mathbb{F}$, (C2), and (P) for the counts. Supporting theorems: `qBinomial_eval`, `qBinomial_expansion`, `gaussBinom_mul_qDen`, `card_subspaces_eq_gaussBinom`, `qBinomial_expansion_count` and `subCount_pascal`. The subspace-count part (`spanMap`, `fibreEquiv`, `card_li_eq`, `gaussBinom_mul_liProd`) is shared with our package for 00000002490 (same author).

Axiom audit. `lake build` succeeds. `#print axioms` reports [`propext`, `Classical.choice`, `Quot.sound`] for `conjecture_2478`, `qBinomial_expansion`, `qBinomial_expansion_count`, `gaussBinom_mul_qDen` and `card_subspaces_eq_gaussBinom`. The file contains no `sorry` and no `native_decide`, and it adds no axioms.

Related package. Conjecture 00000002490 is about the subspace count itself. Its package proves (C1), both q-Pascal rules for the counts, and the interval closure of the subspace lattice. This package is about the expansion (E) and its counting interpretation (C2).

Sources

[W] Wikipedia, “Gaussian binomial coefficient”, https://en.wikipedia.org/wiki/Gaussian_binomial_coefficient (retrieved 2026-10-05). Section “q-binomial theorem”: “There is an analog of the binomial theorem for q-binomial coefficients, known as the Cauchy binomial theorem”, followed by the displayed formula (E). The page also gives the product-formula definition and the Pascal analogs, and states that $\binom{n}{k}_q$ “counts the number of k-dimensional vector subspaces of an n-dimensional vector space over” $\mathbb{F}_q$.

Mathlib: `card_linearIndependent`, in `Mathlib/LinearAlgebra/Matrix/GeneralLinearGroup/Card.lean`.